

Computational Methods for Single-Cell RNA-Seq Analysis of *Arabidopsis thaliana*

Part 1 — Computational Environment and Quick Start

SingleCell Pipeline

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Materials and Methods

Materials

The pipeline source code is available at <https://github.com/cliford2001/SingleCell>:

```
bash  
git clone https://github.com/cliford2001/SingleCell.git  
cd SingleCell
```

The repository is meant to sit inside a working directory that also holds the raw data and the pipeline's output folder — for instance `~/your-workdir/SingleCell`, with `~/your-workdir/resultados` alongside it. That parent directory, not the repository alone, is what gets mounted into the container, so that code, input matrices, and results are all visible together at `/workspace`.

Docker packages R, Python, and every pipeline dependency into one portable **image** (`matigara/scrnaseq:latest`), so the analysis runs identically on any machine with Docker installed. Two equivalent ways to start it:

Option 1 — single command

```
bash  
docker run -it --rm \  
-v ~/your-workdir:/workspace \  
matigara/scrnaseq:latest /bin/bash
```

Pulls the image if missing and opens a shell. Replace `~/your-workdir` with the actual path to that working directory if it lives somewhere else.

Option 2 — the repository's docker-compose.yml

Run as-is from the repository root:

```
bash  
docker compose pull  
docker compose run --rm r bash
```

`pull` only reads that file and downloads the image it names — unrelated to git, which just placed the file on disk. Same `~/your-workdir` → `/workspace` mount as Option 1.

Either way, `/workspace` **is** `~/your-workdir` on the host: files written there persist after the container exits.

Once inside the shell, Chapter 1-2 commands are typed into R, and Chapter 3 commands into Python:

```
R  
R
```

```
python  
python3
```

Methods

Chapter 1 — Single-cell processing

Running Option 1 opens a self-contained terminal inside the container, separate from the host shell it was launched from:

```
— bash —
docker run -it --rm \
  -v ~/your-workdir:/workspace \
  matigara/scrnaseq:latest /bin/bash
```

R is then started from that terminal, on its own:

```
— R —
R
```

Chapter 1 (`capitulo1_single_cell.R`) runs entirely from this session.

Initialization

After entering the R session, Chapter 1 starts by defining the two paths that control the analysis. `PIPELINE_DIR` points to the helper scripts inside the container. `DATA_DIR` is the mounted project directory, and `base_dir` is the root under which all step-specific outputs are written. In a standard Docker run these paths do not need to be changed.

```
— R —
PIPELINE_DIR <- "/workspace/ScRNASeq-Docker/workflow"
DATA_DIR     <- "/workspace/."
base_dir     <- file.path(DATA_DIR, "resultados")
```

The analysis environment is then initialized by loading the package set, custom plotting utilities, and core workflow functions. A fixed random seed is used throughout the chapter, Seurat 5 compatibility is enabled, and the working directory is set to the mounted project folder.

```
— R —
source(file.path(PIPELINE_DIR, "load_libraries.R"))
source(file.path(PIPELINE_DIR, "custom_seurat.R"))
source(file.path(PIPELINE_DIR, "ScRNA_Analysis_Functions.R"))

set.seed(1807)
options(Seurat.allow.s4 = FALSE)
setwd(DATA_DIR)
```

Finally, the pipeline creates the output directory structure under `resultados/`. The helper functions use `output_dir` as the active destination for plots and tables; it is initialized to `base_dir` and reassigned at the start of later sections.

```
— R —  
list2env(create_pipeline_dirs(base_dir), envir = .GlobalEnv)  
output_dir <- base_dir
```

Section 1 — Data loading and pre-filter QC

Section 1 begins by selecting the first output folder and declaring the sample manifest. Each sample entry provides the CellRanger `filtered_feature_bc_matrix` directory, the label used in figures, and the experimental condition used downstream. The same structure is used for any number of samples; only paths, labels, and condition names need to be adapted.

```
— R —
output_dir <- dir_01

samples <- list(
  list(file = "cellranger/sample_1/outs/filtered_feature_bc_matrix",
        label = "sample_1", condition = "condition_1"),
  list(file = "cellranger/sample_2/outs/filtered_feature_bc_matrix",
        label = "sample_2", condition = "condition_1"),
  list(file = "cellranger/sample_3/outs/filtered_feature_bc_matrix",
        label = "sample_3", condition = "condition_2"),
  list(file = "cellranger/sample_4/outs/filtered_feature_bc_matrix",
        label = "sample_4", condition = "condition_2")
)
```

One color is assigned to each sample label so that all QC and UMAP plots use the same visual identity throughout the workflow.

```
— R —
colors <- c(
  "sample_1" = "#66c2a5",
  "sample_2" = "#41ae76",
  "sample_3" = "#fc8d62",
  "sample_4" = "#e34a33"
)
```

The Arabidopsis mitochondrial and chloroplast gene prefixes are then used to compute organelle-derived read fractions for every cell. These pre-filter QC distributions are saved as `qc_prefilter.pdf` and are inspected before choosing the filtering thresholds in the next step.

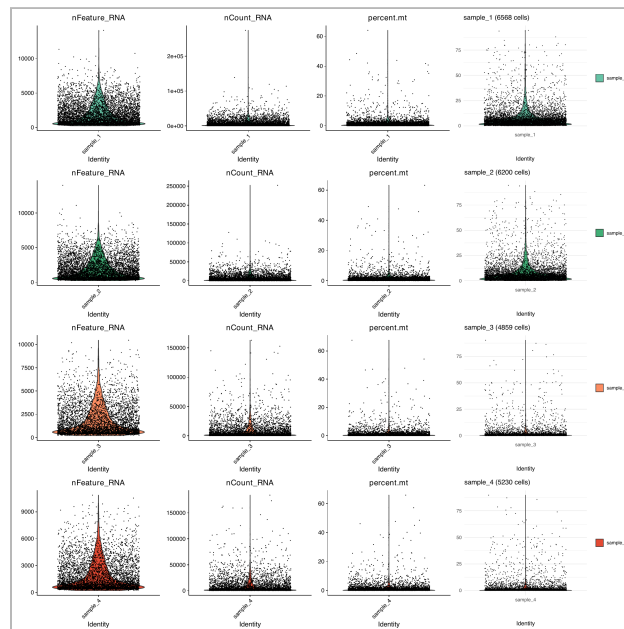
```

# R
mt_pattern <- "^ATMG"
cp_pattern <- "^ATCG"

seurat_list_raw <- load_seurat_samples(
  samples = samples,
  DATA_DIR = DATA_DIR,
  mt_pattern = mt_pattern,
  cp_pattern = cp_pattern
)

plot_qc_batch(seurat_list_raw, colors, "qc_prefilter.pdf")

```



*Representative pre-filter QC output produced by
`plot_qc_batch()`.*

Section 2 — Cell filtering and doublet detection

Section 2 applies the first cell-level exclusion criteria. The default thresholds retain cells with at least 200 detected genes and less than 5% mitochondrial signal. DoubletFinder is then applied sample-by-sample through the filtering helper, and the same QC panel is regenerated after filtering.

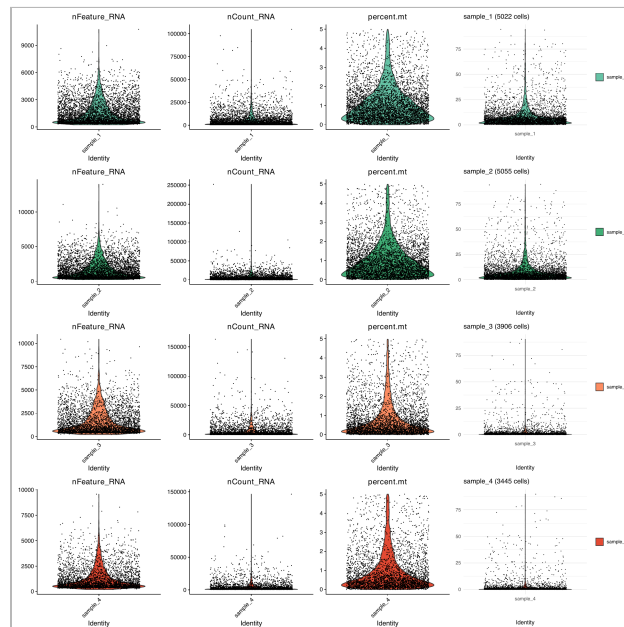
```
R
output_dir <- dir_01

seurat_list <- filter_seurat_samples(
  seurat_list_raw,
  min_features = 200,
  max_mt = 5
)

plot_qc_batch(seurat_list, colors, "qc_postfilter.pdf")
```

The filtered list is saved as a checkpoint so the workflow can restart from this point without reloading and refiltering the raw matrices.

```
R
saveRDS(
  seurat_list,
  file.path(dir_objects, "seurat_list_postfilter.rds")
)
```



Representative post-filter QC output after cell filtering and doublet removal.

Section 3 — Merge and initial preprocessing

The filtered samples are merged into one Seurat object and processed with the standard log-normalization workflow. Variable features are selected by VST, the matrix is scaled, PCA is computed over 30 components, and an initial UMAP is generated from the PCA space. This pre-Harmony UMAP is used to visualize sample structure before batch correction.

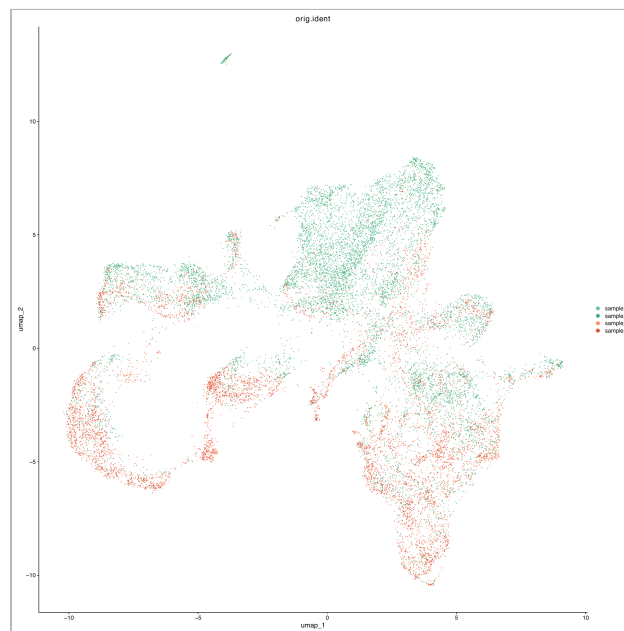
```
— R —
output_dir <- dir_01

pbmc_harmony <- reduce(seurat_list, merge) %>%
  NormalizeData(verbose = FALSE) %>%
  FindVariableFeatures (
    selection.method = "vst",
    nfeatures = 2000,
    verbose = FALSE
  ) %>%
  ScaleData(verbose = FALSE) %>%
  RunPCA(npcs = 30, verbose = FALSE) %>%
  RunUMAP(reduction = "pca", dims = 1:30, verbose = FALSE)
```


The pre-correction UMAP is saved by sample identity, followed by a checkpoint of the merged object.

```
R
save_pdf(
  DimPlot(pbmc_harmony, group.by = "orig.ident", cols = colors),
  "umap_preharmony.pdf"
)

saveRDS(
  pbmc_harmony,
  file.path(dir_objects, "pbmc_harmony_preharmony.rds")
)
```



Representative UMAP before Harmony correction, colored by sample identity.

Section 4 — Harmony batch correction

Harmony is then run on the sample identity field (`orig.ident`) to reduce sample-level batch structure while preserving biological variation. The UMAP embedding is recomputed from the Harmony reduction, and downstream sections use Harmony coordinates rather than the original PCA space.

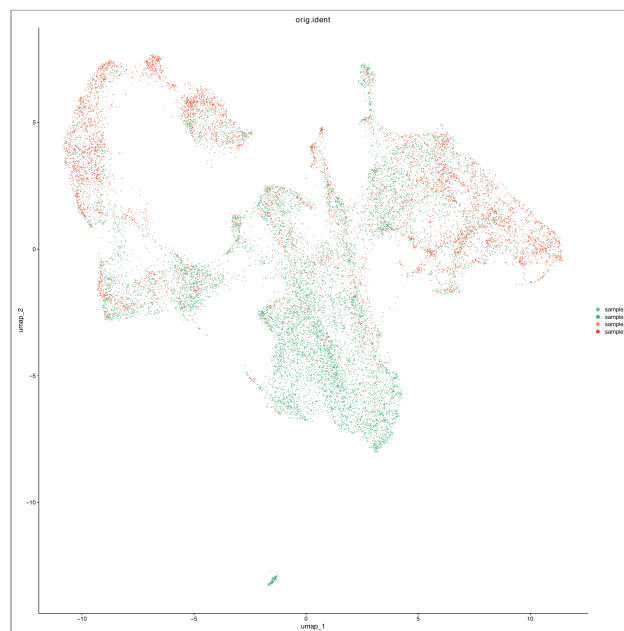
```
R
output_dir <- dir_01

pbmc_harmony <- pbmc_harmony %>%
  RunHarmony("orig.ident", plot_convergence = FALSE) %>%
  RunUMAP(reduction = "harmony", dims = 1:30, verbose = FALSE)
```

The post-Harmony UMAP and object checkpoint are written to disk for later resolution testing, clustering, annotation, and export.

```
R
save_pdf(
  DimPlot(pbmc_harmony, group.by = "orig.ident", cols = colors),
  "umap_postharmony.pdf"
)

saveRDS(
  pbmc_harmony,
  file.path(dir_objects, "pbmc_harmony_postharmony.rds")
)
```



Representative UMAP after Harmony correction, colored by sample identity.